📘 MODUL PRAKTIKUM

IoT Monitoring System (ESP32 + Python + Dashboard) – Windows 11

---

1. 🎯 Tujuan

Mahasiswa mampu:

- Menghubungkan ESP32 ke WiFi
- Mengirim data sensor ke MQTT broker
- Menerima data dengan Python
- Menampilkan dashboard real-time

---

2. 🧱 Arsitektur Sistem

[DHT11 Sensor] → [ESP32] → WiFi → [MQTT Broker (Windows)] → [Python Dashboard]

---

3. 🔧 Kebutuhan

Hardware

- ESP32 DevKit
- Sensor DHT11 / DHT22
- Kabel jumper + breadboard

Software (Windows 11)

- Arduino IDE
- Python 3.10+
- Mosquitto MQTT Broker (Windows)

- Library Python:
    - paho-mqtt
    - streamlit
    - pandas

---

4. ⚙️ Instalasi di Windows 11

4.1 Install Python

- Download dari: https://www.python.org
- ⚠️ Centang: **“Add Python to PATH”**

Cek:

python --version

4.2 Install Mosquitto MQTT Broker

Download:
👉 https://mosquitto.org/download/

Langkah:

1. Install Mosquitto
2. Jalankan service atau manual:

mosquitto -v

Jika error port:

netstat -ano | findstr 1883

4.3 Install Library Python

pip install paho-mqtt streamlit pandas

---

5. 🔌 Wiring ESP32 + DHT11

DHT11	ESP32
VCC	3.3V
GND	GND
DATA	GPIO 4

---

6. 💻 Program ESP32

Gunakan Arduino IDE → pilih board **ESP32 Dev Module**

Install Library:

- WiFi (default)
- PubSubClient
- DHT sensor library

Kode ESP32

```
#include <WiFi.h>
#include <PubSubClient.h>
#include "DHT.h"

#define DHTPIN 4
#define DHTTYPE DHT11

const char* ssid = "WIFI_KAMU";
const char* password = "PASSWORD_WIFI";
const char* mqtt_server = "192.168.1.5"; // IP laptop Windows

WiFiClient espClient;
PubSubClient client(espClient);
DHT dht(DHTPIN, DHTTYPE);

void setup_wifi() {
 WiFi.begin(ssid, password);
```

```
 while (WiFi.status() != WL_CONNECTED) {
   delay(500);
 }
}

void reconnect() {
 while (!client.connected()) {
   if (client.connect("ESP32Client")) break;
   delay(2000);
 }
}

void setup() {
 Serial.begin(115200);
 setup_wifi();
 client.setServer(mqtt_server, 1883);
 dht.begin();
}

void loop() {
 if (!client.connected()) reconnect();
 client.loop();

 float t = dht.readTemperature();
 float h = dht.readHumidity();

 if (!isnan(t) && !isnan(h)) {
   String payload = "{\"temp\":" + String(t) + ",\"hum\":" + String(h) + "}";
   client.publish("iot/sensor", payload.c_str());
 }

 delay(3000);
}
```

---

7. 🖥️ Python Subscriber

File: `subscriber.py`

```
import paho.mqtt.client as mqtt
import json
```

```
def on_message(client, userdata, msg):
  data = json.loads(msg.payload.decode())
  print(f"Suhu: {data['temp']} °C | Kelembaban: {data['hum']} %")

client = mqtt.Client()
client.connect("localhost", 1883)

client.subscribe("iot/sensor")
client.on_message = on_message

client.loop_forever()
```

Jalankan:

```
python subscriber.py
```

---

8. 📊 Dashboard Real-Time (Streamlit)

File: `dashboard.py`

```
import streamlit as st
import paho.mqtt.client as mqtt
import json
import pandas as pd
import time

data_list = []

def on_message(client, userdata, msg):
  global data_list
  data = json.loads(msg.payload.decode())
  data_list.append(data)

client = mqtt.Client()
client.connect("localhost", 1883)
client.subscribe("iot/sensor")
client.on_message = on_message
client.loop_start()

st.title("📊 Dashboard IoT ESP32 (Windows 11)")

chart = st.empty()
```

```
while True:
  if len(data_list) > 0:
     df = pd.DataFrame(data_list)
     chart.line_chart(df)
  time.sleep(2)
```

Jalankan:

streamlit run dashboard.py

➡ Buka browser:

http://localhost:8501

---

9. 🔁 Multi Node (WMN Concept)

Tambahkan beberapa ESP32:

Topic:

iot/node1
iot/node2

Atau gunakan:

- ESP-MESH (WiFi mesh)
- BLE Mesh

---

10. 🧪 Tugas Praktikum

Level 1 (Dasar)

- ESP32 kirim data
- Python menerima data

Level 2 (Menengah)

- Dashboard real-time
- Multi sensor

Level 3 (Lanjut)

- Multi node (3 ESP32)
- Filtering data
- Simpan ke CSV

---

11. 📈 Simpan Data ke CSV

Tambahkan di dashboard:

```
df.to_csv("data_sensor.csv")
```

---

12. 🧠 Integrasi AI (Opsional)

```
from sklearn.linear_model import LinearRegression
import numpy as np

X = np.arange(len(data_list)).reshape(-1,1)
y = np.array([d['temp'] for d in data_list])

model = LinearRegression()
model.fit(X, y)

pred = model.predict([[len(data_list)]])
print("Prediksi suhu:", pred)
```

---

13. ❗ Troubleshooting Windows

ESP32 tidak connect

- Cek WiFi
- Pastikan IP MQTT benar

MQTT tidak jalan

mosquitto -v

Port 1883 bentrok

- Disable service lain

Streamlit tidak jalan

pip install --upgrade streamlit

---

14. 🎯 Output Akhir

Mahasiswa menghasilkan:

- Sistem IoT berbasis ESP32
- Dashboard real-time
- Multi-node monitoring system
- (Opsional) prediksi AI

🚀 Pengembangan Lanjutan

Bisa dikembangkan ke:

- Smart Greenhouse 🌱
- Smart City 🌆
- Monitoring Banjir 🌊
- Industrial IoT 🏭